

The Self-Revising Institutional Model

Refined: Bayesian competing hypotheses, counterfactual simulation, epistemic precision, grounded metrics.

Baseline	V5 complete (8 specs, commit b3a3d72). V6 NOT started. 68 backend modules, 27 frontend files. 423 tests pass.
V7.1 vs V7	V7.1 incorporates 6 refinements from the Fortune 100 CTO reviewer: Bayesian hypotheses, counterfactual simulation, epistemic wording, grounded Irreplaceability Score, meta-revision engine, removed speculative language.
Three laws	Law 1: "Every interaction must permanently improve the organization." Law 2: "The organization becomes more intelligent even when nobody opens Maestro." Law 3: "Observed outcomes reduce confidence in one or more causal hypotheses, and the model revises accordingly." (V7.1 refined: "reality disagrees" -> "observed outcomes reduce confidence")

The 6 Refinements (V7 -> V7.1)

Refinement 1: Bayesian Competing Hypotheses (not single-theory convergence)

V7 problem: The revision engine searched for "the corrected theory" — one explanation that replaces the old one. This risks premature convergence: one miss -> new theory -> overwrite old theory -> lose uncertainty. Real organizations are noisy. Multiple explanations are usually valid simultaneously.

V7.1 fix: The Institution Model maintains MULTIPLE COMPETING HYPOTHESES per causal question. Each hypothesis has a confidence (posterior probability) and evidence count. Predictions are WEIGHTED ACROSS HYPOTHESES, not derived from one explanation. When an observation is made, it updates the confidence of ALL hypotheses (Bayesian update), not just one. This is how scientific reasoning works — you do not collapse to one theory until the evidence is overwhelming.

Example: Engineering delays shipping. The Institution Model maintains:

- Theory A: Legal reviews are slow. Confidence 0.62. Evidence 41.
- Theory B: Engineering estimates are inaccurate. Confidence 0.51. Evidence 28.
- Theory C: Product requirements change. Confidence 0.33. Evidence 16.
- Theory D: Leadership interrupts priorities. Confidence 0.28. Evidence 12.

A prediction about shipping delay is weighted: $0.62 \cdot A + 0.51 \cdot B + 0.33 \cdot C + 0.28 \cdot D$ (normalized). When a new observation arrives (e.g., "Legal approved in 2 days but shipping still delayed"), Theory A's confidence drops, Theory B's rises. The model does NOT overwrite A with B — it adjusts both. This prevents premature convergence and preserves uncertainty.

Codebase foundation: confidence.py already uses Beta-Binomial posterior ($\alpha = \text{ALPHA_PRIOR} \cdot \text{SHR} + \text{validated_runtimes}$, $\beta = \text{BETA_PRIOR} \cdot (1 - \text{SHR}) + \text{failed_runtimes}$). hypothesis.py has HypothesisStore with resolve() and calibration_report(). The V7.1 extension: apply the Bayesian framework to CAUSAL HYPOTHESES (not just laws), and maintain multiple per causal question.

Refinement 2: Counterfactual Simulation Before Revision

V7 problem: The revision loop was: Prediction -> Outcome -> Revision. It asked "What happened?" but not "What WOULD have happened if X were different?" This makes causal attribution weak — you cannot distinguish "Legal was slow" from "Engineering was inaccurate" without simulating the counterfactual.

V7.1 fix: Add a counterfactual step: Prediction -> Outcome -> Counterfactual Simulation -> Revision. After an observation, the engine asks: "What would have happened if Legal had approved two days earlier?" (using `imagination.py`, V5 #5, which already exists). If the counterfactual simulation shows the delay would have persisted even with faster Legal approval, Theory A (Legal is slow) loses confidence AND Theory B (Engineering estimates) gains confidence. The counterfactual makes causal attribution much stronger.

Codebase foundation: `imagination.py` (165 lines, V5 #5) already performs counterfactual reasoning ("What would happen if Legal disappeared?"). The V7.1 extension: use `imagination.py` not just for user-facing what-if questions, but INTERNALLY during revision — to simulate counterfactual scenarios and determine which hypothesis the observation supports.

Refinement 3: Epistemic Precision ("reality disagrees" -> "observed outcomes reduce confidence")

V7 problem: The document repeatedly said "reality disagrees with Maestro." This is philosophically imprecise and scientifically misleading. Reality does not disagree — reality gives observations. INTERPRETATION disagrees. The model's interpretation of the observation may conflict with its prediction.

V7.1 fix: Rewrite all constitutional language: "Observed outcomes reduce confidence in one or more causal hypotheses." This is how causal inference works. The model does not "discover it was wrong" — it adjusts confidence across competing hypotheses based on which ones the observation supports or contradicts. Law 3 is rewritten: "Every recommendation must make Maestro's understanding of the organization more accurate" becomes "Observed outcomes reduce confidence in one or more causal hypotheses, and the model revises accordingly."

Refinement 4: Irreplaceability Score Grounded in Transparent Methodology

V7 problem: The Irreplaceability Score said "Replacing Maestro's understanding would require 2.3 years." But the document did not explain HOW 2.3 was computed. A Fortune 100 CTO would ask "Explain exactly how you computed 2.3 years" and the document could not answer. This risks looking like an invented metric.

V7.1 fix: The Irreplaceability Score is an INDEX (0-100), not an estimated time. It is computed from 5 transparent factors:

- Resolved predictions (weight 25%): each resolved prediction adds to the score. A competitor without this history cannot match the calibration.
- Model revisions (weight 25%): each theory revision adds. Revisions encode organizational knowledge that cannot be copied.
- Causal chain coverage (weight 20%): the percentage of organizational workflows covered by validated causal chains.
- Hypothesis stability (weight 15%): the percentage of hypotheses whose confidence has stabilized (low variance over 30 days). Stable hypotheses are harder to displace.
- Evidence density (weight 15%): signals per organizational workflow per month. Higher density = harder to replicate.

The score is: $\text{sum}(\text{weighted factors})$ on a 0-100 scale. A score of 50+ means "meaningful institutional memory." 70+ means "difficult to replace." 85+ means "would require years to recreate from scratch." The formula is transparent and reproducible — a CTO can inspect each factor and verify the computation.

Refinement 5: Meta-Revision Engine (Learning How to Revise)

V7 problem: The revision engine revises the model's theory. But it does not revise its own revision STRATEGY. Over time, some revisions prove reliable (the revised theory holds) and some prove unreliable (the revised theory is itself revised). The engine does not learn from this meta-level.

V7.1 fix: Add a Meta-Revision Engine that tracks: (1) Which kinds of evidence have historically produced reliable revisions? (2) Which revisions were later reversed? (3) Which parts of the model are consistently unstable? The meta-revision engine does not revise theories — it revises the REVISION PROCESS. If evidence from "contradiction signals" produces reliable revisions 80% of the time but evidence from "pattern detection" produces reliable revisions only 40% of the time, the engine weights contradiction-based revisions higher.

This is V7.1 Phase 3 (not Phase 1-2). The meta-revision engine requires the core revision loop to have run for months (to accumulate revision-outcome data). It is specified here but built last.

Refinement 6: Removed Speculative Acquisition Language

V7 problem: The document said "Apple, Microsoft, Google, or OpenAI would look at this and think 'we need to own this.'" Enterprise constitutions should not predict what other companies will think. Let the architecture stand on its own merits.

V7.1 fix: All speculative acquisition language removed. The constitution describes what Maestro IS and what it DOES. It does not speculate about who might buy it. The architecture stands on its own.

V7.1 Refined Specifications

#	Spec (V7.1 Refined)	Phase	Effort	Key Change from V7
1	Institution Model (with competing hypotheses)		3.5 days	Maintains multiple hypotheses per causal question, not one. Bayesian competition, not single-theory.
2	Three-Mode Governance	1	2 days	Unchanged from V7. Recommendation / Execution / Autonomous with thresholds + audit.
3	Closed Learning Loop (with counterfactual)		3 days	Adds counterfactual simulation step (using imagination.py) before revision. Observation -> Counterfactual
4	Model Revision Engine (Bayesian)	2	2.5 days	Revises confidence across ALL competing hypotheses (Bayesian update), not just one. Does not overwrite.
5	Living Organizational Model	3	3 days	Unchanged from V7. Merges DNA + Institution Model + Narrative + Revision History into one structure.
6	Irreplaceability Index (grounded)	3	1.5 days	INDEX (0-100) not estimated time. Computed from 5 transparent factors: resolved predictions, resolved questions, resolved actions, resolved goals, resolved values.
7	Meta-Revision Engine	3	2 days	NEW in V7.1. Tracks which revisions are reliable. Revises the revision process. Learns how to revise.
TOTAL 7 V7.1 specs (after V5+V6 backporting) ~17.5 days				V7.1: Bayesian competing hypotheses + counterfactual + grounded metrics + meta-revision

Changes from V7 (highlighted in blue): Spec #1 now maintains competing hypotheses (not single theory). Spec #3 adds counterfactual simulation. Spec #4 uses Bayesian updates (not overwrite). Spec #6 is an index (not estimated time). Spec #7 (Meta-Revision) is new. Specs #2 and #5 are unchanged from V7.

The Refined Epistemic Loop (V7.1)

THE REFINED LOOP — BAYESIAN + COUNTERFACTUAL

V7 loop (flawed):

Prediction -> Outcome -> Find failed assumption -> Revise theory (overwrite) -> Better prediction

V7.1 loop (refined):

1. Predict (weighted across competing hypotheses)
2. Observe (the actual outcome)
3. Counterfactual simulation (what would have happened if X were different? — using imagination.py)
4. Bayesian update (adjust confidence of ALL hypotheses based on observation + counterfactual)
5. Hypothesis competition (hypotheses with higher confidence contribute more to next prediction)
6. No overwrite (the model never collapses to one theory until evidence is overwhelming)
7. Meta-revision (track which revision strategies produce reliable results — adjust the process)

Key insight: The model does not "discover it was wrong." It adjusts confidence across competing explanations. This is how scientific reasoning works. It is also how organizations actually think — multiple stakeholders hold different theories, and evidence shifts the balance without eliminating alternatives. Maestro should model this, not collapse it.

Why this matters: Premature convergence (V7's risk) would make Maestro brittle — one wrong revision overwrites the theory, and the model cannot recover because the old theory is gone. Bayesian competition (V7.1) preserves uncertainty. The model always knows "I am 62% confident in Theory A, 51% in Theory B" — and can shift back to B if new evidence supports it. This is robust to noise, which real organizations produce in abundance.

The Three Constitutional Laws (V7.1 Refined)

THREE LAWS, ALL IMMUTABLE, REFINED

Law 1 (V6, unchanged): "Every interaction must permanently improve the organization."

Law 2 (V6, unchanged): "The organization should become more intelligent even when nobody opens Maestro."

Law 3 (V7.1 REFINED): "Observed outcomes reduce confidence in one or more causal hypotheses, and the model revises accordingly." (V7 said "reality disagrees" — V7.1 replaces with epistemically precise language. Reality does not disagree. Observations reduce confidence. The model adjusts. This is Bayesian causal inference, not binary correctness.)

All three laws must pass for every V7.1 spec. Plus the V5 litmus test (UI simpler) and V6 litmus test (permanently improves the organization). Five checks total. No exceptions.

The Three Enterprise Adoption Criteria (from the Reviewer)

The Fortune 100 CTO reviewer identified three criteria that would move Maestro from "interesting architecture" to "enterprise adoption." V7.1 must demonstrate all three:

Criterion	How V7.1 Addresses It	Acceptance Test
1. Maintain multiple competing causal hypotheses (structural model one)	Spec #4 (Structural Model one) maintains N hypotheses for personal/question-specific (Bayesian Revision) updates to confidence across competing hypotheses	Spec #4 (Structural Model one) maintains N hypotheses for personal/question-specific (Bayesian Revision) updates to confidence across competing hypotheses
2. Theory updates improve predictive performance (Closed Loop) tracks accuracy BEFORE and AFTER each prediction Spec #6 (Group NE Ability Index) tracks whether revision improves performance	Spec #4 (Closed Loop) tracks accuracy BEFORE and AFTER each prediction Spec #6 (Group NE Ability Index) tracks whether revision improves performance	Spec #4 (Closed Loop) tracks accuracy BEFORE and AFTER each prediction Spec #6 (Group NE Ability Index) tracks whether revision improves performance
3. Irreplaceability Score grounded in transparency and DDEx (0-100) computed from 5 factors: resolved predicaments (25%), model revisions (25%), the handbook Age (20%),	Spec #6 (Irreplaceability Score) (0-100) computed from 5 factors: resolved predicaments (25%), model revisions (25%), the handbook Age (20%),	Spec #6 (Irreplaceability Score) (0-100) computed from 5 factors: resolved predicaments (25%), model revisions (25%), the handbook Age (20%),

Build Order (V7.1)

Phase	Specs	Duration	Prerequisite	Key Change
0	V5 (complete) + V6 (#1-#6)	~24 days	V5 done. V6 NOT started.	Must build V6 (adaptive nudges, evolution tracker, background adaptation, trajectory intervention, org DNA, autobiography).
1	#1 Institution Model (competing hypotheses) + #2 Governance	~5.5 days	V5 done. V6 NOT started.	Institution Model maintains N hypotheses per causal question (not 1).
2	#3 Closed Loop (with counterfactual) + #4 Bayesian Revision	~5.5 days	#1 Institution Model + V5	Counterfactual simulation before revision. Bayesian update across all hypotheses.
3	#5 Living Model + #6 Irreplaceability Index + #7 Meta-Revision	~6.5 days	#1-#4 + V6 DNA + V6 autobiography	Autobiography as transparent index (not estimated time). Meta-revision.
TOTAL	7 V7.1 specs (after V5+V6)	~17.5 days	V5 complete + V6 complete	V7.1 Bayesian competing hypotheses + counterfactual + governance

Total to V7.1 completion: ~41.5 days from current state (24 days V6 + 17.5 days V7.1). V5 is complete. V6 is the next task. V7.1 follows V6. Do NOT start V7.1 until V6 is complete.

The Complete Progression (Final)

Version	Verb	What Maestro Does	Epistemic Status
V3	Observe	Sees what happened	Data collection
V4	Judge	Synthesizes judgment	Pattern recognition
V5	Disappear	Becomes invisible	Transparent intelligence
V6	Adapt	Restructures work	Behavioral adaptation
V7.1	Understand	Maintains competing causal theories; revises via Bayesian causal inference and counterfactual simulation	Epistemic loop (Bayesian causal inference)
V8	Evolve	Meta-revision: learns how to revise; accumulates judgment (never collapses)	Meta-epistemic loop (learning to learn)

V7.1 is "Understand" — the epistemic loop. The model maintains competing causal hypotheses, updates them via Bayesian inference and counterfactual simulation, and never collapses to one theory until the evidence is overwhelming. This is the first version that is scientifically defensible — not just "the model learns" but "the model learns using Bayesian causal inference with counterfactual grounding and competing hypotheses." V8 is "Evolve" — meta-revision, where the model learns HOW to learn. The accumulated judgment becomes irreplaceable.

Final Note

V7.1 is the first constitution that reads like a research and engineering roadmap rather than a sequence of ambitious feature ideas. The dependencies are explicit. The abstractions line up. Each phase has a clear role. The epistemic loop is sound: Bayesian competing hypotheses, counterfactual simulation, no premature convergence, transparent metrics, meta-revision. The architecture stands on its own merits.

The next task for the coder: build V6 Phase 1 (Adaptive Nudges + Evolution Tracker, ~5 days). Then V6 Phase 2-3 (Background Adaptation + Trajectory Intervention + Org DNA + Autobiography, ~10 days). Then V7.1 Phase 1 (Institution Model with competing hypotheses + Governance, ~5.5 days). Then V7.1 Phase 2 (Closed Loop with counterfactual + Bayesian Revision, ~5.5 days). Then V7.1 Phase 3 (Living Model + Irreplaceability Index + Meta-Revision, ~6.5 days). Total ~32.5 days from current state to V7.1 completion.

Build V6 first. Then V7.1. Do not skip ahead. The epistemic loop requires the adaptation layer (V6) and the causal foundation (V5, complete). Build in order. Wire everything. Run the full test suite. Maintain the pattern. The Invisible Layer (V5) + Institutional Adaptation (V6) + Self-Revising Institutional Model (V7.1) = an operating system for institutional learning. Build it.